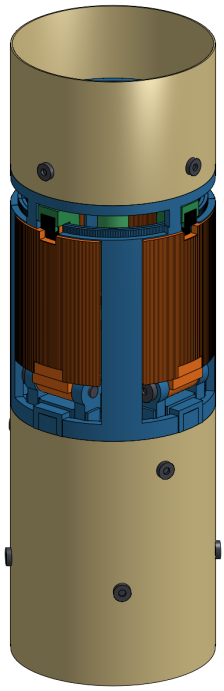
V1 – Screwed assembly guide

ARC BT80 airbrake, *reinforced bottom spring, 90° stop* – your Assembly 23 model with every bonded joint turned into a screw joint.

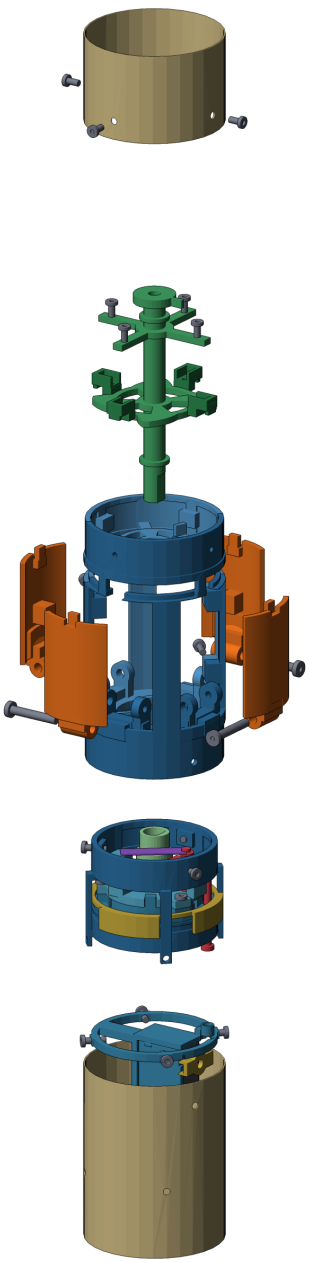
Onshape document (public, Free plan): ARC BT80 Airbrake - V1 Screwed + V2 Compact (Claude build)
Elements: **V1-A** closed state, **V1-B** fully open (90° / rotor 65° / trigger up), **ASM-V1** assembly with mates and hardware, **SAMPLES** unloaded spirals, **01 - V1 screwed + V2 compact (source)** Feature Studio (parameter *V2 compact head* = off).



V1 closed (Onshape render). Tan = reference BT80 tubes with the drill holes; dark grey = hardware.

1 What changed from your model

Joint (your model)	Now
C1 / C2 housing seam: bond	4 × M3x8 across the seam, on inward ribs at z=8 and z=44 (peg alignment kept)
Hinge pin H + end cap K: printed pin, bond	M4x25 button-head screw per hinge; thread into the +Y ear; bores are now 4.5 mm; K removed
Catch cap L: bond to rotor arm (could not slide on: closed socket)	Open-top clip with a screw boss under the arm; 1 × M3x8 each. Crown arm gets a 3.4 mm hole
Sleeves: glue to BT80	Upper sleeve: 4 × M3x6 into pads. Lower sleeve + dock ring + tube: 3 × M3x10
Cradle S to dock feet: bond	4 × M3x10 radial through tube, foot, cradle pad (foot extended 1.5 mm lower, pad thickened to R26)
Lid F to casing M: bond	3 × M3x10 vertical through lid ears into casing bosses
Q collar, G keeper, W return-leaf root	Still bonded: sections are under 3 mm, no room for a screw
Seam boss z 63..67	Raised to z 63.5..67 so the new cap screw heads clear it at every rotor angle



Exploded overview. Order top to bottom: upper tube + sleeve screws, rotor with caps and cap screws, fins with hinge screws around the housing, cartridge, cradle, lower tube.

2 Print list

STLs are in `onshape/exports/v1/print`. Each file is auto-oriented (least overhang of the 6 axis orientations) with z-min at 0; check orientation in your slicer. The 2.6 mm pilot holes and 3.4 mm clearance holes are printed as modelled: do not scale them, and do not add supports inside them.

File	Qty	Part	Notes
C1_housing_half_1_x1	1	C1 housing half (carries fins 1 and 4, the 4 pilot holes of the seam)	Pegs/sockets align it with C2. Print upright as exported.
C2_housing_half_2_x1	1	C2 housing half (carries fins 2 and 3, seam counterbores)	Mirror of C1 with pegs.
P_fin_x4	4	Fin (cam-driven panel, 90° stop)	Identical x4; hinge bore is 4.5 mm.
L_catch_cap_x4	4	Catch cap (open-top clip with screw boss)	Identical x4. Print the boss up so the 2.6 mm pilot is vertical.
R_rotor_x1	1	Rotor: shaft, 4 cams, crown arms, knob	Four 3.4 mm vertical holes in the crown arms.
U_output_hub_x1	1	Cartridge output hub	
Q_thrust_collar_x1	1	Lower thrust collar	Stays bonded (2.5 mm tall ring).
X_double_d_adapter_x1	1	Double-D adapter	
T_winding_cap_x1	1	Winding and latch cap	
D_cartridge_dock_x1	1	Cartridge dock (3 ring pads, 4 feet)	Feet have 3.4 mm radial holes.
N_bayonet_collar_x1	1	Bayonet retaining collar	
I_locking_pin_x1	1	Quarter-turn locking pin	
M_casing_x1	1	Cartridge casing (3 lid bosses)	
F_lid_x1	1	Cartridge lid (3 screw ears)	
A_pushrod_x1	1	Short side pushrod	
B_release_yoke_x1	1	Upward release yoke	
S_servo_cradle_x1	1	SG90 cradle (4 pads, 2.6 mm pilots)	
G_servo_keeper_x1	1	Servo keeper	Stays bonded (1.5 mm plate).
J_horn_saddle_x1	1	Double-arm horn saddle	Measure your horn first (as in your model).
samples/S_unloaded_spiral_*_x1	1 each	Unloaded spiral springs 1.6 / 2.0 / 2.4 mm beam	Print references from your Part Studio 24, unchanged.
samples/W_unloaded_return_leaf_x1	1	Return leaf (1 mm installation preload)	Root stays bonded.

3 Hardware and tools

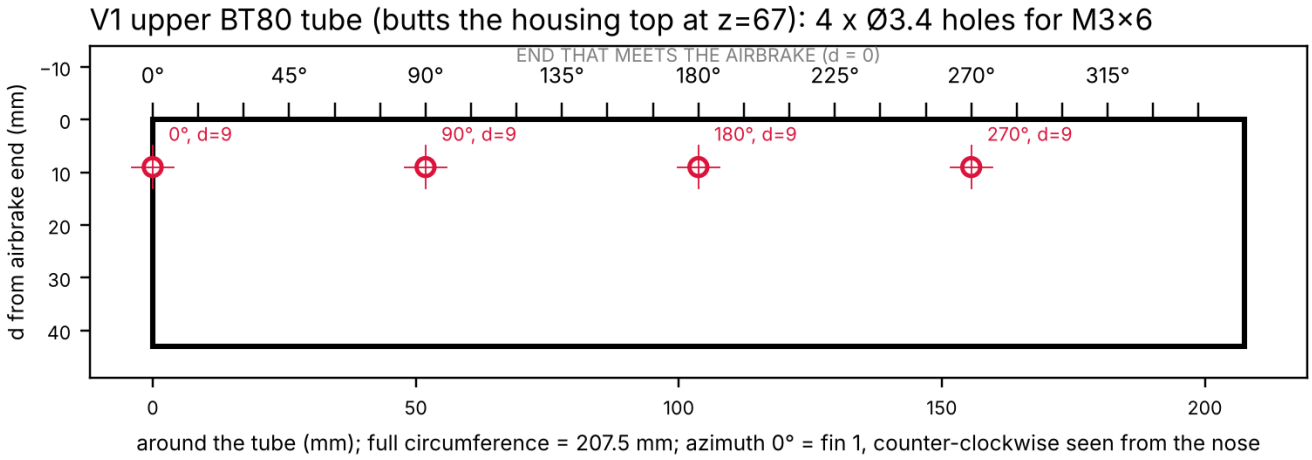
Ref in CAD	Fastener	Qty	Joint	Driver
HW seam screw 1-4	M3 x 8 socket-head cap (ISO 4762)	4	C2 counterbore → thread into C1 across the seam; 2 per seam at z=8 and z=44	2.5 mm hex
HW L1-L4 cap screw	M3 x 8 button head (ISO 7380)	4	Rotor arm → catch cap boss (vertical)	2.0 mm hex
HW H1-H4 hinge screw	M4 x 25 button head (ISO 7380)	4	Through -Y ear and fin barrel, threads into +Y ear (replaces printed pin H and cap K)	2.5 mm hex
HW upper sleeve screw 1-4	M3 x 6 button head	4	Upper BT80 tube → pads behind the upper sleeve (z=76)	2.0 mm hex
HW dock ring screw 1-3	M3 x 10 button head	3	Lower BT80 tube → housing sleeve → dock ring pad (z=-26.5)	2.0 mm hex
HW cradle screw 1-4	M3 x 10 button head	4	Lower BT80 tube → dock foot → cradle pad (z=-67.2)	2.0 mm hex
HW lid screw 1-3	M3 x 10 button head	3	Lid ear → casing boss (vertical)	2.0 mm hex

Totals: M3x6 × 4, M3x8 × 8 (4 socket + 4 button), M3x10 × 10, M4x25 × 4 = 26 screws. Not printed or screwed: 1 SG90 servo with horn, 2 BT80 airframe tubes, epoxy for Q, G, the return-leaf root and the horn saddle J.

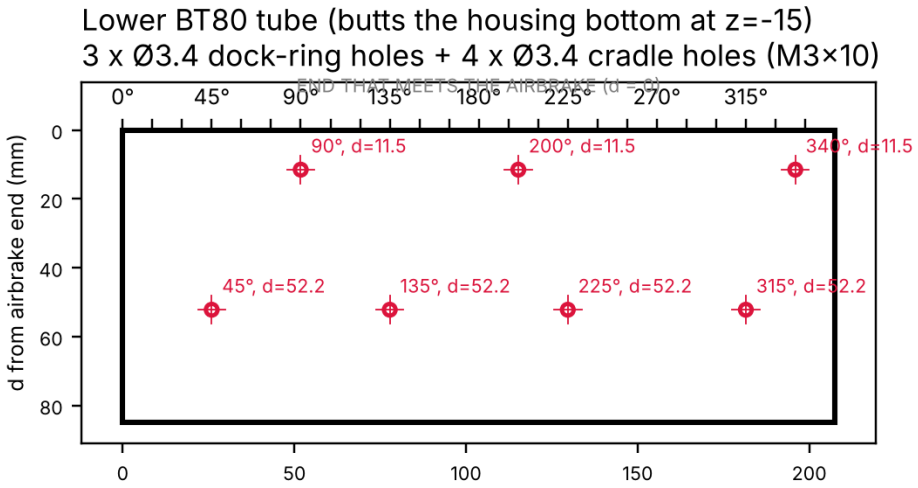
- Self-tapping into PLA/PETG: run each screw once into its pilot hole before final assembly (this cuts the thread), then back it out.
- Snug plus about 1/8 turn. Over-tightening strips a printed thread long before the screw yields; M3 into a 2.6 mm pilot gives a few N·m of strip torque at best.
- The tube screws only pass through the 0.5 mm tube wall and thread into printed pads. Use a drop of thread-lock only on steel-to-steel joints (none here).

4 Prepare the BT80 tubes

BT80 is OD 66.04 mm, ID about 65.0 mm. The sleeves are 64.6 mm OD. Azimuth 0° is fin 1; angles run counter-clockwise looking down the rocket from the nose. Distances d are measured from the tube end that meets the airbrake. Drill 3.4 mm holes.



Upper tube template (unrolled, 1:1 horizontal scale).



Lower tube template. Mark directly, or drill through the printed sleeve holes while the tube is on.

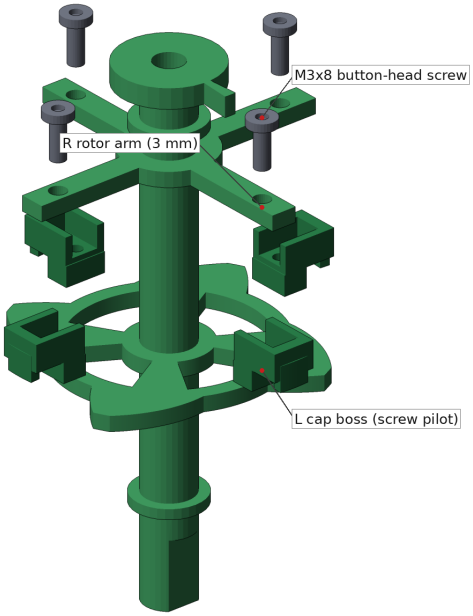
Tube	Hole	d (mm)	Azimuth	Screw
Upper	4 holes	9.0	0, 90, 180, 270°	M3x6
Lower	3 holes	11.5	90, 200, 340°	M3x10 (dock ring)
Lower	4 holes	52.2	45, 135, 225, 315°	M3x10 (cradle)

The lower tube needs at least 85 mm of length below the housing to cover the cradle; the upper tube needs at least 43 mm above the housing (the reference tubes in the CAD are 85 and 43 mm).

5 Assembly sequence

Step 1 – Rotor group (catch caps)

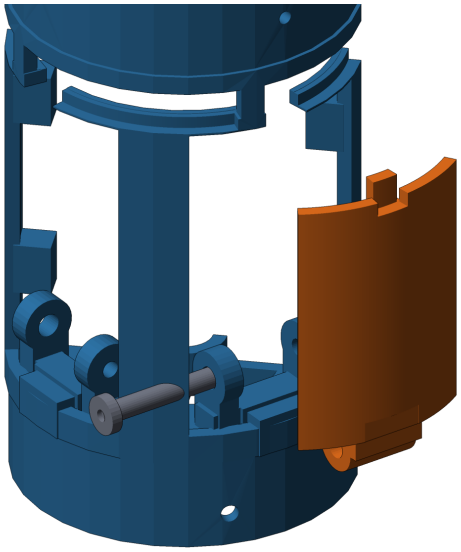
Hold the rotor with the arms at 0/90/180/270°. Offer each catch cap **from below**: the 3 mm arm sits in the open-top slot, the hook points outward, the boss with the 2.6 mm pilot is directly under the arm. Drive one M3x8 button-head screw per cap from above through the arm hole into the boss. Do this with the housing halves still apart: the closed housing leaves no access.



Caps L1–L4 offered up under the arms, screws above. The rotor arms carry the screw holes.

Step 2 – Fins on the housing halves

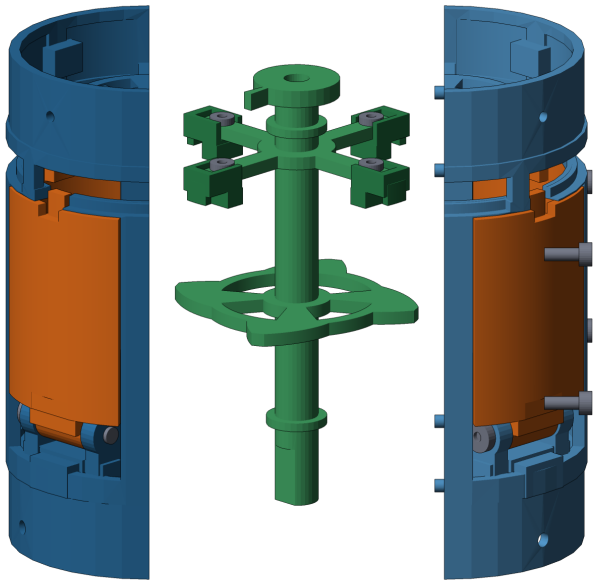
C1 carries fins 1 and 4, C2 carries fins 2 and 3 (fin n sits at (n–1) × 90°). For each fin, seat the barrel between the two ears and run an M4x25 button-head screw through the ear on the **clockwise side** (seen from the nose), through the barrel, and into the other ear. The head sits in the pocket; stop when it seats. The fin must swing 0–90° with no drag. If it binds, back the screw off a quarter turn and re-check that the barrel sits centred between the ears (0.4 mm each side).



Hinge detail: M4x25 screw in line with the ear bores; fin barrel goes between the ears.

Step 3 – Close the housing around the rotor

Lay the rotor with its caps between the open halves, drop its shaft into the two bearing bores, and close the halves; the two pegs on C2 locate in the sockets of C1. Fit the four **M3x8 socket-head seam screws** (2.5 mm hex): heads go into the counterbores on the C2 ribs and thread into C1. Swing **fin 2 open** to reach the two screws at the 45° seam and **fin 3 open** for the two at the 225° seam. I ray-checked the key path in the open-fin CAD state: all four screw heads are reachable by a straight 2.5 mm hex key (V1-B). Turn the rotor through 0–65° afterwards; it must move freely.

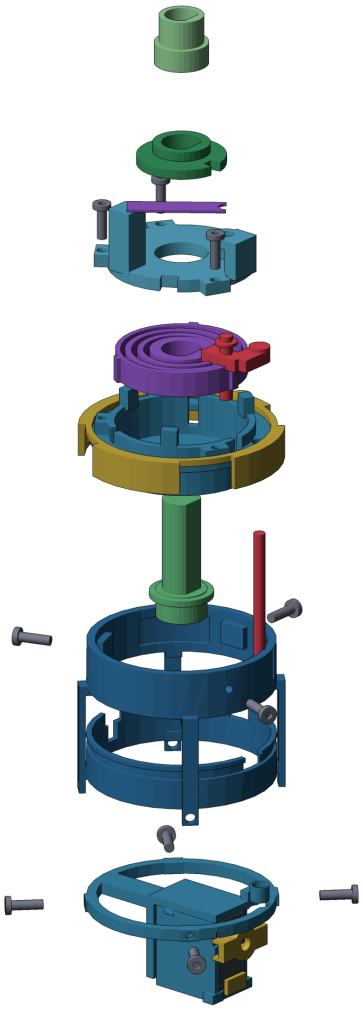


Halves apart: C1 (left) with fins 1 and 4, C2 (right) with fins 2 and 3, rotor and caps between, seam screws out to the right.

Step 4 – Cartridge and servo cradle (bench sub-assemblies)

The cartridge mechanism is unchanged from your design. The order below follows your part labels (“fit last”, “attach after lid”, “bond after insertion”) and the geometry; I have not run the mechanism physically.

- **Cradle S:** slide the SG90 into the cradle, bond keeper G, fit horn saddle J over the horn.
- **Casing M:** insert hub U from the top; bond Q to the hub from below after insertion; place the printed spiral on the D stem with its three tabs in the casing slots.
- **Lid F:** add the lid and fasten with three M3x10 button-head screws through the lid ears into the casing bosses (ears at 145°, 215° and 335°, chosen to stay clear of the pushrod, latch guide, return leaf and locking pin).
- **Then** winding cap T, yoke B with pushrod A, return leaf W (root bonded), and the double-D adapter X last.
- **Dock D:** seat the casing in the dock and lock it with bayonet collar N and pin I (quarter turn, unchanged).
- **Cradle to dock:** the cradle pads sit against the dock feet. The four M3x10 screws go in through the lower tube in Step 5.



Cartridge exploded top to bottom: adapter X, cap T, lid F with its three screws, spring V, casing M, hub U and collar Q, dock D with ring screws, cradle S with the four cradle screws.

Step 5 – Join module, housing and lower tube

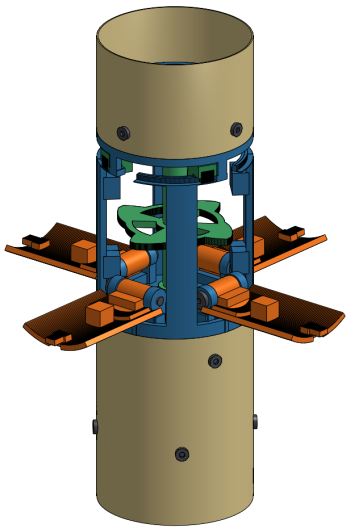
Push rotor D-key into the hub socket while sliding the dock ring into the lower sleeve (0.5 mm radial gap). Insert this assembly, module end first, into the top of the lower BT80 tube until the housing base ring seats on the tube end. Align the tube holes with the sleeve holes and fit **3 × M3x10** (dock ring, 90/200/340°) and **4 × M3x10** (cradle pads, 45/135/225/315°). Each of the 3 ring screws passes through the tube wall, the sleeve (3.4 mm clearance) and threads into a pad inside the dock ring; the cradle screws pass through the tube and the dock foot into the cradle pad.

Step 6 – Upper tube

Slide the upper tube down over the upper sleeve until it butts the housing top (z=67). Fit 4 × M3x6 button-head screws (0/90/180/270°, 9 mm from the tube end) into the pads behind the sleeve.

Step 7 – Function check

- Each fin swings 0–90° and stops hard at 90° on the outer compression stop.
- Rotor turns 0–65°; at 65° the catch hooks clear the fin tabs and the cams kick the fins open.
- Trigger lifts 4.5 mm and releases the latch; the return leaf brings it back.
- All 26 screws are snug; no screw head proud of the tube except the button heads (1.65 mm).



V1-B: all fins at 90°, rotor at 65°, trigger up (Onshape render).

6 Checks run on this model

Check	Result
Feature regeneration (Part Studio + FeatureScript)	OK, about 3–8 s
V1-A closed	56 parts, all watertight; printed-part interference: none; screw/thread engagement pairs: 26 (intended).
V1-B open (90°, 65°, 4.5 mm)	56 parts, all watertight; printed-part interference: none; screw/thread engagement pairs: 26 (intended).
Interference known from your original	Return leaf W reference vs yoke B in the pressed-trigger state (10 mm³); latch disk T vs yoke B at some rotor angles; both are in your model and unchanged
Seam screw access	Straight 2.5 mm hex key reaches all four heads with fins 2 and 3 open (ray test)
Assembly mates	24 features (3 mate groups, 4 hinge revolutes 0–90°, rotor revolute 0–65°, trigger slider 0–4.5 mm, servo-pusher revolute): all solve OK

Not verified: physical print fit and thread strength, servo torque and latch behaviour (unchanged from your design), and the mate motion itself (the Onshape API accepted the mates but would not move them from a script, so open the assembly and drag a fin to confirm).